Dev Sanjay Sakaria

Robotics & Electromechanical Systems Engineer · Medical Devices

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SUMMARY

Robotics engineer focused on **medical devices and surgical robotics**, with hands-on experience taking electromechanical systems from concept to tested prototype. At IIT Madras Research Park I developed the force-control and haptic teleoperation stack for a 7-DOF robot-assisted ultrasound system and built the test phantoms used to validate it. My background spans control (admittance/impedance, haptics), mechanical design and fabrication, sensing and perception, and system architecture with risk analysis for regulated devices. Based in Bengaluru and looking for R&D roles in medtech and surgical robotics.

EXPERIENCE

Robotics Engineer, Robot-Assisted Tele-Ultrasound · Healthcare Technology Innovation Centre (HTIC), IIT Madras Research Park — Jan 2026 – Sep 2026
Chennai, India

- Developed an **admittance-based force control framework** for a 7-DOF manipulator, giving compliant, stable probe–tissue contact for robot-assisted ultrasonography.
- Designed and implemented a **real-time external control layer** on top of the robot's internal controller, improving force tracking and responsiveness for scanning tasks.
- Architected and prototyped a **grounded haptic master interface** with impedance-based force feedback for intuitive, safe teleoperation.
- Modelled system dynamics and tuned controller parameters for stability and safe human–robot interaction in a clinical setting.
- Designed and built a **pneumatically actuated breathing phantom**, taking it from a first prototype to a robot-ready test bed that reproduces respiratory motion for validating contact control on a moving body.

Robotics Systems Design Intern · Hourglass Medical (M.Eng Capstone, University of Illinois) — May 2025 – Aug 2025
Champaign, IL, USA

- Led the **system architecture** of a hands-free robotic surgical lighting system, controlled by smart glasses with a jaw-clench sensor and laser pointer, using **MBSE in SysML** (block definition and internal block diagrams) and the V-model.
- Defined the six subsystems and their interfaces (glasses, perception, lighting planner, control, light arms, safety), including laser-spot 3D localization, IK-based obstacle-aware motion planning and CAN-based actuation over ROS 2.
- Performed a high-level **FMEA** and designed the safety system (emergency stop, no-go zones, IR proximity sensing, manual override).

Robotics Intern, Multi-Robot Localization · Robert Bosch Centre for Cyber-Physical Systems, Indian Institute of Science (IISc) — Jan 2024 – Jun 2024
Bengaluru, India

- Built multi-robot localization pipelines in ROS that fuse fixed infrastructure-camera poses with on-board LiDAR, IMU, wheel odometry and depth camera in an **EKF**, feeding RTAB-Map SLAM.
- Integrated keypoint-based (YOLOv8-pose + PnP) **6D pose estimation** with uncertainty into the localization stack.
- Improved localization accuracy by about **39%** through systematic tuning and validation in Gazebo, Isaac Sim and on the Volta AMR fleet.

Automation Intern · Sansera Engineering — Dec 2022 – Feb 2023
Bengaluru, India

- Designed an automated bearing-assembly station, developed its PLC logic and HMI, and integrated an inline vision-inspection system.

EDUCATION

M.Eng. Autonomy & Robotics · University of Illinois Urbana-Champaign, Grainger College of Engineering · GPA: 3.71/4.00 — Aug 2024 – Dec 2025

- Coursework: Surgical Technologies, Soft Robotics, Whole Body Musculoskeletal Biomechanics, Robotics, Principles of Safe Autonomy, Autonomous Vehicle System Engineering, Applied Machine Learning

B.Tech. Mechatronics Engineering · Manipal Institute of Technology Jun 2020 – Jul 2024

- Coursework: Mechanical Design, Control Systems, Hydraulics & Pneumatics, Digital Signal Processing, Artificial Intelligence

SELECTED PROJECTS

- **Soft Robotic Toe Separator · Best Design Process Award**: Pneumatically actuated silicone (Dragon Skin / Ecoflex) PneuNet toe separator for post-operative bunion recovery, with air chambers formed by a dissolvable sugar-insert casting method. Took part in every stage, from research and requirements through mould design, multi-material casting, pneumatic assembly and on-foot testing.
- **Biomechanics of Taekwondo Punches**: Motion capture and force-plate study of dominant vs non-dominant punches, with OpenSim inverse kinematics and inverse dynamics of the hip, knee and ankle.
- **LiDAR–Camera Calibration & GPS-Free Localization**: Calibration pipeline (93.8% IoU alignment) and ICP map-based localization for a full-size autonomous vehicle; designed the first version of the estimator.
- **Lane Following & Sign Detection**: Camera-based lane detection and Pure Pursuit control in ROS on the GEM e4 vehicle, with YOLO STOP/SLOW sign detection.

SKILLS

Control & Robotics	Admittance / impedance control, haptic teleoperation, force control, kinematics & dynamics modelling, controller tuning, ROS / ROS 2
Medical Device Development	System architecture (MBSE, SysML), FMEA and ISO 14971 risk management, tissue phantoms and test fixtures, biomechanics (OpenSim, motion capture, force plates)
Mechanical Design & Prototyping	SolidWorks, 3D printing, silicone moulding and casting, pneumatic actuation, soft robotics
Electronics & Embedded	Sensors and actuators, CAN bus, microcontrollers, PLC / HMI
Perception & Estimation	OpenCV, YOLO, PnP pose estimation, EKF sensor fusion, LiDAR–camera calibration
Software & Simulation	Python, C++, MATLAB / Simulink, Gazebo, Isaac Sim, MuJoCo